

10 Photosynthesis

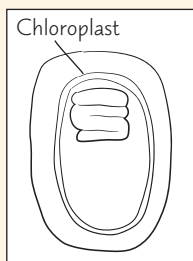
KEY CONCEPTS

- 10.1** Photosynthesis feeds the biosphere *p. 188*
- 10.2** Photosynthesis converts light energy to the chemical energy of food *p. 189*
- 10.3** The light reactions convert solar energy to the chemical energy of ATP and NADPH *p. 192*
- 10.4** The Calvin cycle uses the chemical energy of ATP and NADPH to reduce CO_2 to sugar *p. 201*
- 10.5** Alternative mechanisms of carbon fixation have evolved in hot, arid climates *p. 203*
- 10.6** Photosynthesis is essential for life on Earth: a review *p. 206*

Study Tip

Make a visual study guide: Draw a cell with a large chloroplast, labeling the parts of the chloroplast. As you go through the chapter, add key reactions

for each stage of photosynthesis, linking the stages together. Label the carbon molecule(s) with the most energy and the carbon molecule(s) with the least energy.



Go to Mastering Biology

For Students (in eText and Study Area)

- Get Ready for Chapter 10
- BioFlix® Animation: Photosynthesis
- Figure 10.13 Walkthrough: How Linear Electron Flow During the Light Reactions Generates ATP and NADPH

For Instructors to Assign (in Item Library)

- Activity: The Light Reactions
- BioFlix® Tutorial: Photosynthesis

Ready-to-Go Teaching Module

- (in Instructor Resources)
- The Light Reactions (Concept 10.3)



Figure 10.1 Each leaf of this tree is harvesting energy from sunlight and using it to convert CO_2 and H_2O into chemical energy stored in sugar and other organic molecules. The tree uses these sugars for energy and to build its own trunk, branches, and leaves. Remarkably, enough sugars are left over to feed other organisms (like moth larvae, shown below) that cannot carry out this extraordinary transformation.

How do photosynthetic cells use light to change carbon dioxide and water into organic molecules and oxygen?

